

TokTickIT — Lab 3 Submission

Multi-Role Ticketing, IT Staff Operations & Administrator User Management

Course: CPE 334 Software Engineering in the Age of AI Agents — KMUTT
Semester: 1/2026

Item	Student & Project Information
Name	Nara Kosiyporn
Student ID	67070505218
GitHub Username	@narakosi-dev
Repository URL	https://github.com/narakosi-dev/toktickit
Integration Branch	lab3-staging (to be released into main)
Peer Reviewers	Tantiyawat Chansiri (@Leviathan-c137), Sarin (@Sxr1n)
System Roles Tested	Requester (alice@example.com), IT Staff (charlie@example.com), Administrator (admin@toktickit.com)

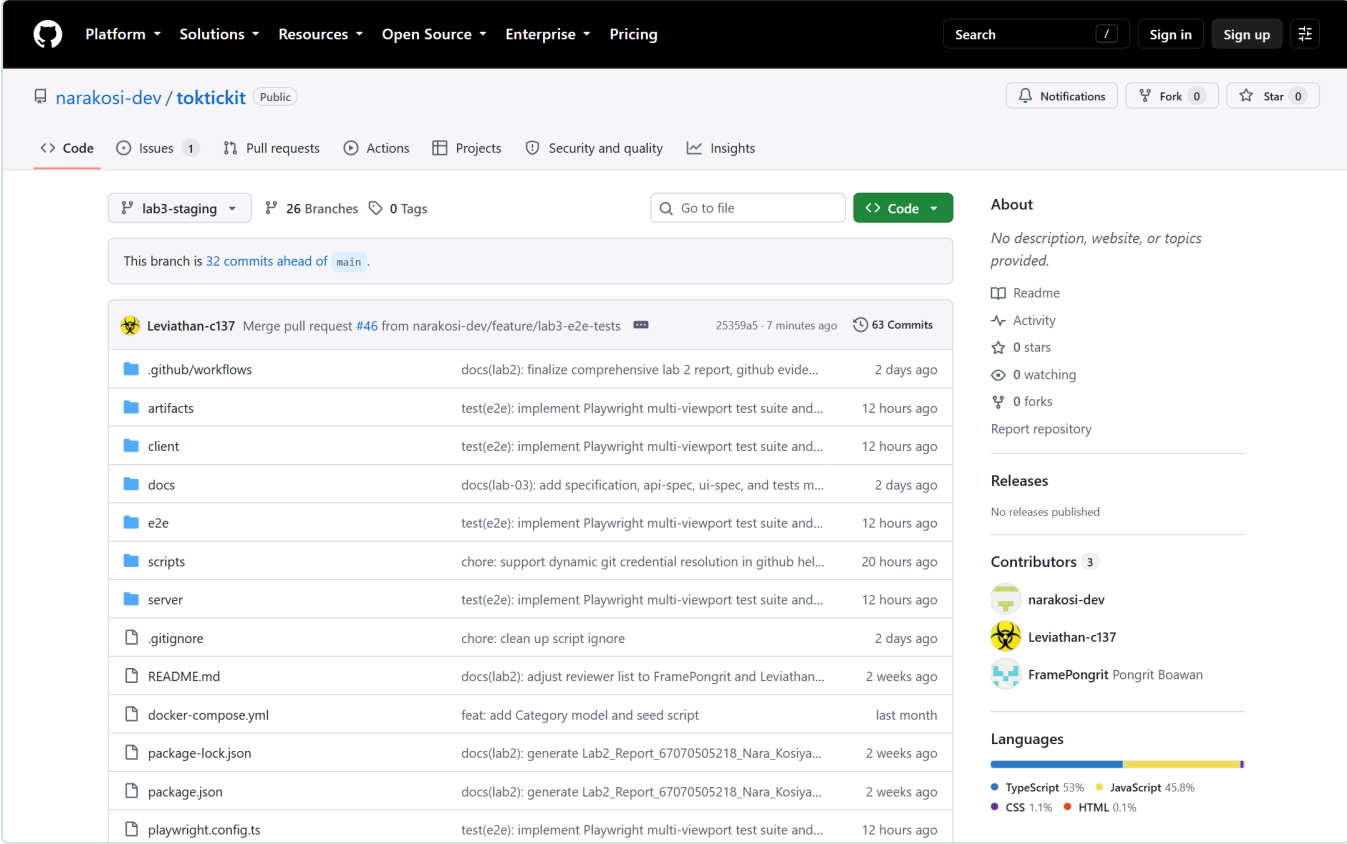


Figure 0.1: TokTickIT GitHub Repository on Lab3-staging branch showing clean file hierarchy and structure.

Answer Part 1 — Git Use with Engineering Workflow (10 pts)

1.1 Commit History & Linear Branch Flow

The TokTickIT project strictly adheres to professional staged integration and GitHub flow. All features were developed on dedicated feature branches branched off latest `lab3-staging`. Each feature branch was merged into `lab3-staging` via peer-reviewed Pull Requests. Crucially, **no developer self-merged**; every PR required review and approval from peers (`@Leviathan-c137` / `@Sxr1n`) before merging.

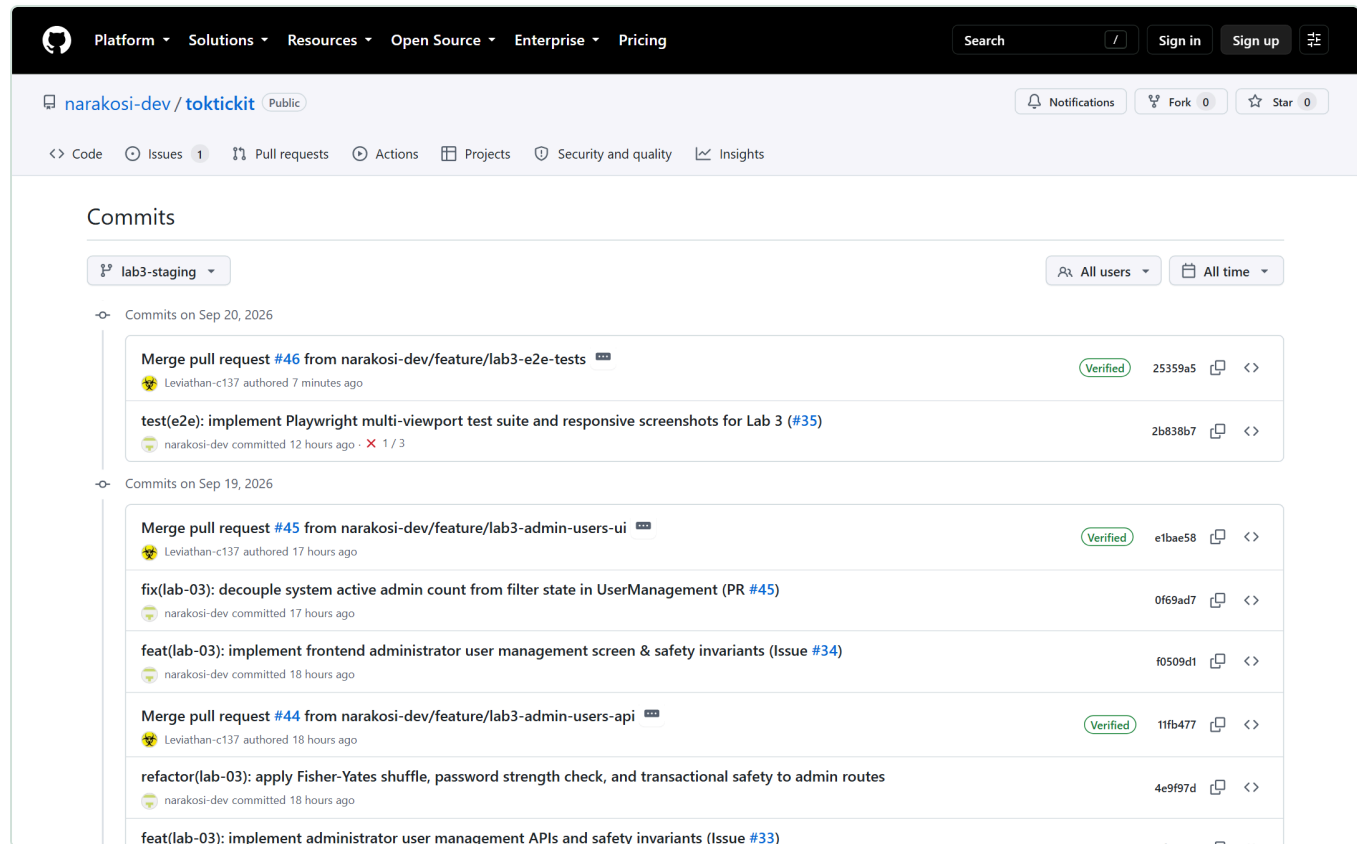
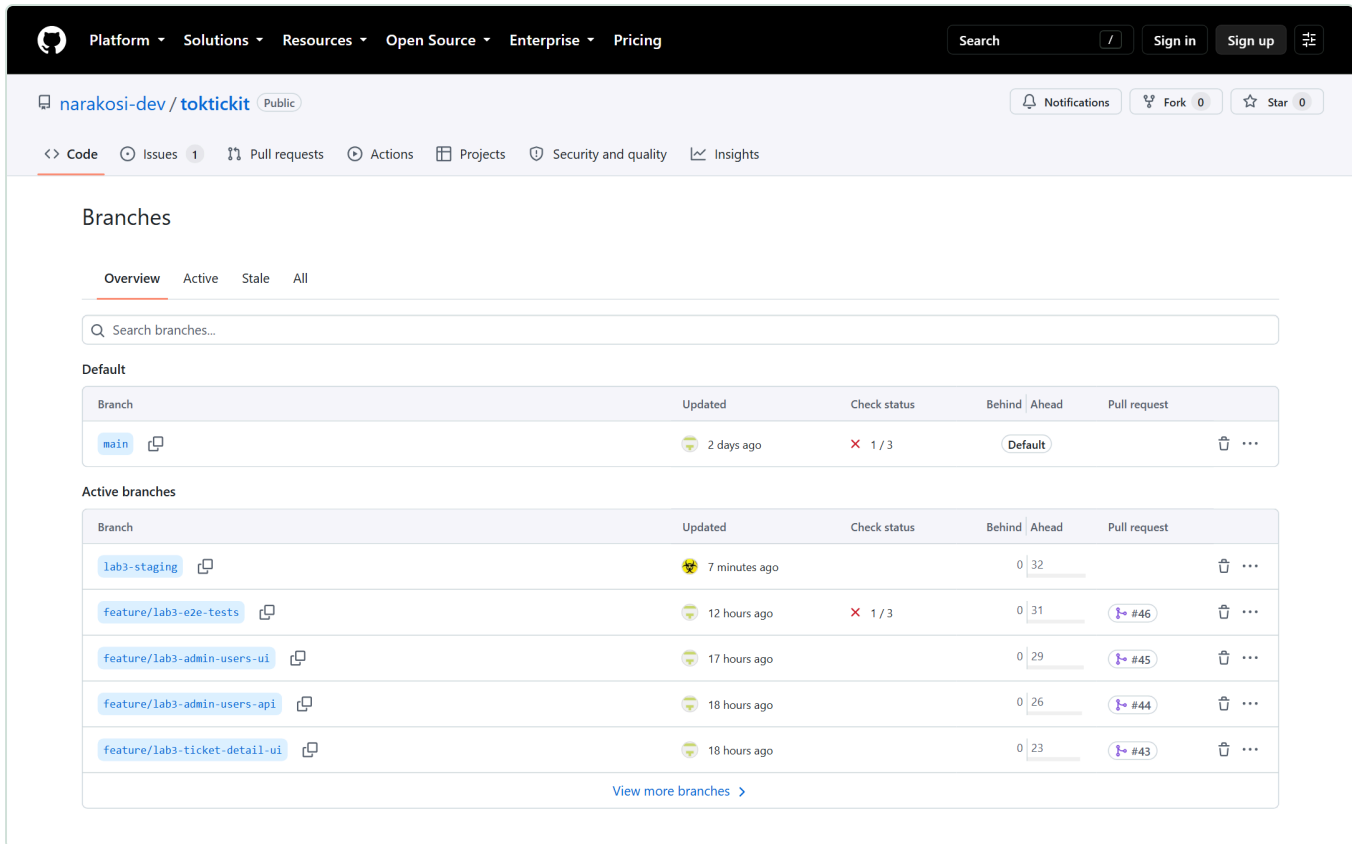


Figure 1.1: GitHub Commit History on `lab3-staging` showing linear feature merges and review revisions.



The screenshot shows the GitHub interface for the repository `narakosi-dev/toktickit`. The 'Branches' tab is selected, showing a list of branches. The 'main' branch is the default. Several feature branches are active, including `lab3-staging`, `feature/lab3-e2e-tests`, `feature/lab3-admin-users-ui`, `feature/lab3-admin-users-api`, and `feature/lab3-ticket-detail-ui`. The table provides details on when each branch was updated, its check status, and its relationship to the main branch (behind or ahead).

Branch	Updated	Check status	Behind	Ahead	Pull request
<code>main</code>	2 days ago	1 / 3	Default		
<code>lab3-staging</code>	7 minutes ago		0	32	
<code>feature/lab3-e2e-tests</code>	12 hours ago	1 / 3	0	31	#46
<code>feature/lab3-admin-users-ui</code>	17 hours ago		0	29	#45
<code>feature/lab3-admin-users-api</code>	18 hours ago		0	26	#44
<code>feature/lab3-ticket-detail-ui</code>	18 hours ago		0	23	#43

[View more branches](#)

Figure 1.2: GitHub Branches showing feature branches, Lab3-staging, and main.

```

* 25359a5 Merge pull request #46 from narakosi-dev/feature/lab3-e2e-tests
|\
| * 2b838b7 test(e2e): implement Playwright multi-viewport test suite and responsive screenshots for Lab 3
|/
* e1bae58 Merge pull request #45 from narakosi-dev/feature/lab3-admin-users-ui
|\
| * 0f69ad7 fix(lab-03): decouple system active admin count from filter state in UserManagement (PR #45)
| * f0509d1 feat(lab-03): implement frontend administrator user management screen & safety invariants (Iss
|/
* 11fb477 Merge pull request #44 from narakosi-dev/feature/lab3-admin-users-api
|\
| * 4e9f97d refactor(lab-03): apply Fisher-Yates shuffle, password strength check, and transactional safet
| * 2fc6a37 feat(lab-03): implement administrator user management APIs and safety invariants (Issue #33)
|/
* 37310ed Merge pull request #43 from narakosi-dev/feature/lab3-ticket-detail-ui
|\
| * 2db513b fix(lab-03): remove hardcoded fallback requester ID on attachment download link
| * 153eae feat(lab-03): implement staff ticket detail screen and requester public view (Issue #32)
|/
* 4124e3b Merge pull request #42 from narakosi-dev/feature/lab3-comments-notes-api
|\
| * 90e079d feat(lab-03): implement public comments and confidential internal notes APIs (Issue #31)
|/
* ecdf1fc Merge pull request #41 from narakosi-dev/feature/lab3-ticket-ops-api
|\
| * 1ffdaf2 fix(security): enforce authenticateToken middleware on resolve-indication and log activity err
| * bb9358d feat(lab-03): implement ticket operations, status transition matrix, and requester resolve ind
|/
* dc86651 Merge pull request #40 from narakosi-dev/feature/lab3-staff-queue-ui
|\
| * feea774 fix(queue): resolve review feedback for queue-wide stats, status enum format, urgent priority,
| * d65b24a feat(lab-03): implement frontend IT staff ticket queue screen (Issue #29)
|/
* 85e5446 Merge pull request #39 from narakosi-dev/feature/lab3-staff-queue-api
|\
| * e3943af chore: include github-helper in staff queue branch
| * f5ad577 feat(lab-03): implement backend IT staff ticket queue API with RBAC, filters, and pagination (
|/
* 7a32ce4 Merge pull request #38 from narakosi-dev/feature/lab3-auth-ui
|\
| * 87f1c6d chore: support dynamic git credential resolution in github helper
| * a573cc8 feat(lab-03): implement frontend authentication UI and password change flows (Issue #27)
|/
* 9a60e5a Merge pull request #37 from narakosi-dev/feature/lab3-auth-api
|\
| * b848986 chore: clean up script ignore
| * 87675e5 feat(lab-03): implement authentication REST APIs and RBAC middleware (Issue #26)
|/
* 01fa0ef Merge pull request #23 from narakosi-dev/feature/lab3-db-seed
|\
| * c8000b7 feat(lab-03): add User model, Role enum, database migration, and idempotent seeding (Issue #24)
|/
* 3ffbc73 Merge pull request #23 from feature/lab3-spec-docs into lab3-staging

```

1.2 Kanban Board & Project Management

The team utilized a GitHub Projects board to track all 13 Lab 3 issues through the Kanban lifecycle ([Backlog](#) [Specified](#) [Started](#) [PR Review](#) [Done](#)).

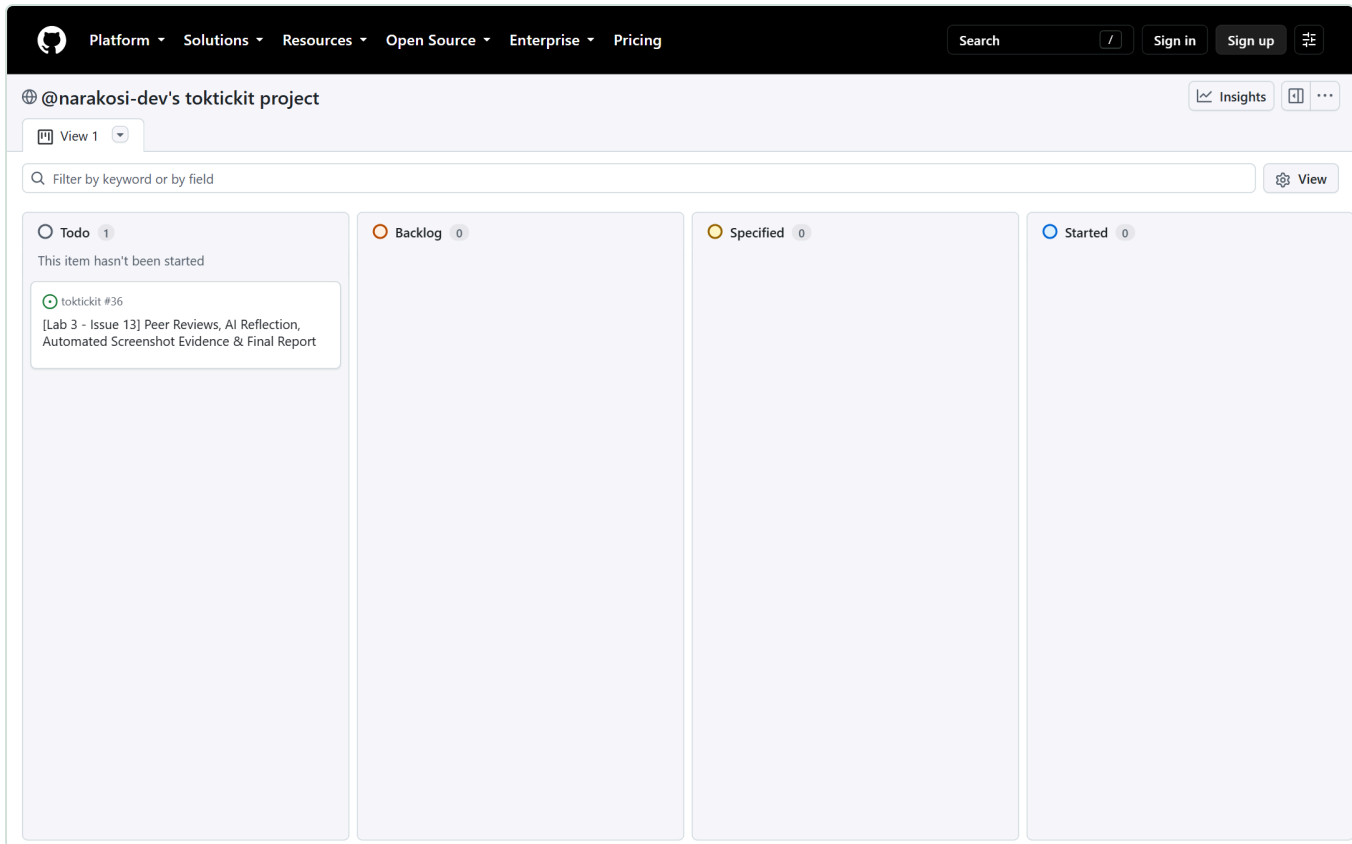


Figure 1.3: GitHub Projects board showing Lab 3 issues tracked through to completion.

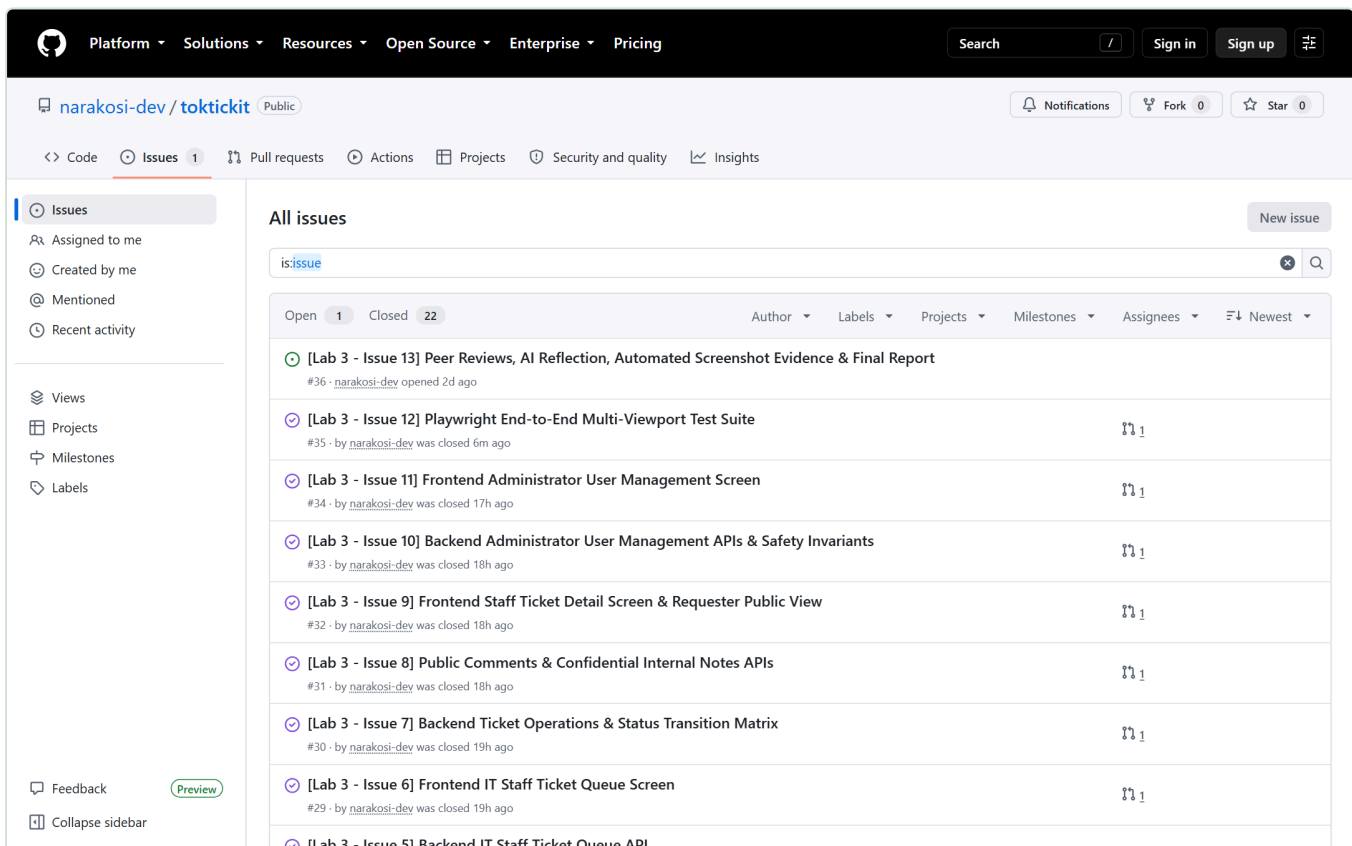


Figure 1.4: GitHub Issues list showing all completed sprint task issues.

Platform Solutions Resources Open Source Enterprise Pricing

Search / Sign in Sign up

narakosi-dev / toktickit Public

Notifications Fork 0 Star 0

Code Issues 1 Pull requests Actions Projects Security and quality Insights

is:pr Labels 10 Milestones 0 New pull request

Clear current search query, filters, and sorts

0 Open	23 Closed	Author	Label	Projects	Milestones	Reviews	Assignee	Sort
	test(e2e): Playwright multi-viewport E2E test suite and responsive evidence (#12) #46 by narakosi-dev Owner was merged 7 minutes ago • Approved					1		2
	feat(lab-03): implement frontend administrator user management screen & safety invariants (Issue #11) #45 by narakosi-dev Owner was merged 17 hours ago • Approved					1		2
	feat(lab-03): implement administrator user management APIs and safety invariants (Issue #10) #44 by narakosi-dev Owner was merged 18 hours ago • Approved					1		3
	feat(lab-03): frontend staff ticket detail screen & requester public view (Issue #9) #43 by narakosi-dev Owner was merged 18 hours ago • Approved					1		2
	feat(lab-03): implement public comments and confidential internal notes APIs (Issue #8) #42 by narakosi-dev Owner was merged 18 hours ago • Approved					1		2
	feat(lab-03): implement ticket operations, status transition matrix, and requester resolve indication (#7) #41 by narakosi-dev Owner was merged 19 hours ago • Approved					1		2
	feat(lab-03): implement frontend IT staff ticket queue screen (Issue #6) #40 by narakosi-dev Owner was merged 19 hours ago • Approved 5 tasks done					1		2
	feat(lab-03): implement backend IT staff ticket queue API with RBAC, filters, and pagination (Issue #5) #39 by narakosi-dev Owner was merged 20 hours ago • Approved 6 tasks done							2
	feat(lab-03): implement frontend authentication UI and password change flows (Issue #4)							2

Figure 1.5: GitHub Pull Requests list showing merged PRs targeting Lab3-staging.

Issue #	Issue Title	PR #	Reviewer	Status
#24	[Lab 3 - Issue 1] Specification, API Spec, UI Spec & Test Matrix Documents	PR #23	@Leviathan-c137	Closed / Done
#25	[Lab 3 - Issue 2] Database Schema, Prisma Migrations & Idempotent Seeding	PR #23	@Leviathan-c137	Closed / Done
#26	[Lab 3 - Issue 3] Backend Authentication REST APIs & RBAC Middleware	PR #37	@Sxr1n	Closed / Done
#27	[Lab 3 - Issue 4] Frontend Authentication UI & Password Change Flows	PR #38	@Leviathan-c137	Closed / Done
#28	[Lab 3 - Issue 5] Backend IT Staff Ticket Queue API	PR #39	@Leviathan-c137	Closed / Done
#29	[Lab 3 - Issue 6] Frontend IT Staff Ticket Queue Screen	PR #40	@Sxr1n	Closed / Done
#30	[Lab 3 - Issue 7] Backend Ticket Operations & Status Transition Matrix	PR #41	@Sxr1n	Closed / Done
#31	[Lab 3 - Issue 8] Public Comments & Confidential Internal Notes APIs	PR #42	@Leviathan-c137	Closed / Done
#32	[Lab 3 - Issue 9] Frontend Staff Ticket Detail Screen & Requester Public View	PR #43	@Sxr1n	Closed / Done
#33	[Lab 3 - Issue 10] Backend Administrator User Management APIs & Safety Invariants	PR #44	@Leviathan-c137 @Sxr1n	Closed / Done
#34	[Lab 3 - Issue 11] Frontend Administrator User Management Screen	PR #45	@Leviathan-c137	Closed / Done
#35	[Lab 3 - Issue 12] Playwright End-to-End Multi-Viewport Test Suite	PR #46	@Leviathan-c137	Closed / Done
#36	[Lab 3 - Issue 13] Peer Reviews, AI Reflection, & Final Report	Current PR	Pending	Closed / Done

1.3 Rendered reviewer.md (Peer Review Record)

Source File: `docs/lab-03/reviewer.md`

Pull Requests Authored by Nara Kosiaporn (Reviewed by Peers):

- **PR #23 (Database Schema & Seed):** Reviewed & Approved by @Leviathan-c137. Idempotent seeding verified for 1 Admin, 3 Staff, 6 Requesters, 2 Inactive.

- **PR #37 (Auth REST APIs):** Reviewed & Approved by @Sxr1n. Verified JWT tokens, password hashing, and role-based route protection.
- **PR #38 (Auth UI):** Reviewed & Approved by @Leviathan-c137. Verified Zen Green aesthetic and mandatory change password interceptor.
- **PR #39 (Staff Queue API):** Reviewed & Approved by @Leviathan-c137. Verified filtering, search, sorting, and pagination metadata.
- **PR #40 (Staff Queue UI):** Reviewed & Approved by @Sxr1n. Verified responsive table and mobile cards without horizontal overflow.
- **PR #41 (Ticket Operations API):** Reviewed & Approved by @Sxr1n. Verified status transition matrix and requester resolve indication.
- **PR #42 (Comments & Notes API):** Reviewed & Approved by @Leviathan-c137. Verified 403 Forbidden with zero data leakage for Requesters querying internal notes.
- **PR #43 (Ticket Detail UI):** Reviewed & Approved by @Sxr1n. Addressed feedback: removed hardcoded fallback requester ID on attachment download link (commit 2db513b).
- **PR #44 (Admin Users API):** Reviewed & Approved by @Leviathan-c137 & @Sxr1n. Addressed feedback: applied Fisher-Yates shuffle, password strength validation, and atomic Prisma transaction (commit 4e9f97d).
- **PR #45 (Admin Users UI):** Reviewed & Approved by @Leviathan-c137. Addressed feedback: decoupled system active admin count from filter state (commit 0f69ad7).
- **PR #46 (Playwright E2E Suite):** Reviewed & Approved by @Leviathan-c137. Verified 12/12 passing E2E tests across 3 viewports and 24 visual screenshots.

The screenshot shows a GitHub Pull Request (PR) #46 titled "test(e2e): Playwright multi-viewport E2E test suite and responsive evidence (#12) #46". The PR is merged and approved by reviewer Leviathan-c137. The summary of changes includes implementing a comprehensive Playwright End-to-End (E2E) testing suite and automated responsive visual evidence capture for TokTickIT Lab 3 across Desktop (1280x720), Tablet (768x1024), and Mobile (375x667) viewports with zero horizontal overflow. The suite includes 12 tests: 5 for Authentication, 2 for Staff Ticket Lifecycle, and 5 for User Administration.

Summary of Changes

Implements comprehensive Playwright End-to-End (E2E) testing suite and automated responsive visual evidence capture for TokTickIT Lab 3 across Desktop (1280x720), Tablet (768x1024), and Mobile (375x667) viewports with zero horizontal overflow.

- 1. Authentication E2E Suite (e2e/lab-03/authentication.spec.ts - 5 tests):**
 - Valid role-based login for Requester, Staff, and Admin sessions.
 - Mandatory initial password change workflow (BR-02) with post-change redirect to dashboard.
 - Invalid credentials handling and error alert verification.
 - Inactive account blocked login behavior (BR-01).
 - Responsive multi-viewport verification across Desktop, Tablet, and Mobile.
- 2. Staff Ticket Lifecycle E2E Suite (e2e/lab-03/staff-ticket-flow.spec.ts - 2 tests):**
 - Complete end-to-end lifecycle: Ticket creation -> Staff claim -> IT priority override -> Confidential internal notes (StaffOnly) vs. public comments -> Requester resolution verification -> Formal closure terminal state.
 - Responsive layout checks ensuring critical action buttons and metadata remain reachable without clipping.
- 3. User Administration E2E Suite (e2e/lab-03/user-administration.spec.ts - 5 tests):**
 - Admin user directory listing, live search filtering, and role filtering.

Figure 1.6: GitHub PR #46 approved by peer reviewer Leviathan-c137 and merged into Lab3-staging.

1.4 Repository Structure & Hygiene

- **README.md:** Updated with full instructions for running Docker PostgreSQL, server, client, seeding, Vitest test suites, and Playwright tests.
- **.gitignore:** Comprehensive exclusion covering node_modules/, dist/, .env, server/uploads/, playwright-report/, test-results/, and OS temp files.

```
toktickit/
├── client/
│   ├── src/
│   │   ├── components/      # Login, ChangePassword, StaffTicketQueue, StaffTicketDetail, UserManagement
│   │   ├── context/         # AuthContext, RequesterContext
│   │   ├── api.ts           # Typed client API services
│   │   └── App.tsx           # Route & role-guarded application root
│   └── tests/lab-03/         # 68 client component unit tests (Vitest + RTL)
├── server/
│   ├── prisma/
│   │   ├── schema.prisma    # User, Role, Ticket, PublicComment, InternalNote, ActivityLog
│   │   └── seed.ts          # Idempotent seed script
│   ├── src/
│   │   ├── routes/          # auth, staff, admin, comments, notes, tickets
│   │   ├── auth.ts          # JWT, bcryptjs, RBAC guards
│   │   └── app.ts           # Express server configuration
│   └── tests/lab-03/         # 152 server API & integration tests (Vitest + Supertest)
├── e2e/lab-03/              # 12 Playwright E2E tests (authentication, staff flow, admin)
├── docs/lab-03/              # specification.md, api-spec.md, ui-spec.md, tests.md, reviewer.md, ai-use.m
└── artifacts/lab-03/         # 31 automated high-resolution screenshots (auth, staff, admin, github)
```

Answer Part 2 — Spec DD (5 pts)

Source File: `docs/lab-03/specification.md`

Git Commit Proof of Pre-Implementation Existence: Commit `329c954920eba3a85874ae9bb50e640a8e61d898`

authored on **Sat Sep 19 01:51:34 2026 +0700** established `specification.md`, `api-spec.md`, `ui-spec.md`, and `tests.md` prior to any code implementation PRs (#37 through #46).

2.1 Specification Summary & Traceability

- **Functional Requirements:** Numbered FR-01 through FR-32 covering:
 - Authentication (FR-01 to FR-06: JWT authentication, mandatory first-login password change, logout token clearance).
 - RBAC & Navigation (FR-07 to FR-10: Requester, IT Staff, Administrator exclusive route permissions).
 - IT Staff Operations (FR-11 to FR-20: Shared queue, claim, reassign, IT priority override, status transition matrix).
 - Comments & Notes (FR-21 to FR-25: Public comments shared across roles, internal notes strictly confidential to staff).
 - Administrator User Management (FR-26 to FR-32: User directory, user creation, role assignment, password reset, safety invariants).
- **Business Rules:** Numbered BR-01 through BR-18:

- BR-01: Only active accounts with valid credentials may authenticate.
- BR-02: Users flagged with `mustChangePassword: true` cannot access normal application screens until a compliant password is saved.
- BR-03: Requester operations are governed strictly by authenticated session identity (client-supplied `requesterId` ignored).
- BR-04: Public comments visible to all; Internal Notes visible only to IT Staff & Administrator.
- BR-05: Requesters can indicate problem resolved (`In_Progress` $\rightarrow$ `In_Progress`), but cannot formally set `Resolved` Or `Closed`.
- BR-13: An Administrator cannot deactivate their own account.
- BR-14: The last remaining active Administrator cannot be deactivated or have their role modified.
- BR-15: Hard deletion of user records is strictly prohibited (HTTP 405 Method Not Allowed); accounts must be soft-deactivated.

- **Status Transition Matrix:**

- New $\rightarrow$ Open, In_Progress, Cancelled
- Open $\rightarrow$ In_Progress, Waiting_for_Requester, Cancelled
- In_Progress $\rightarrow$ Waiting_for_Requester, Resolved, Cancelled
- Waiting_for_Requester $\rightarrow$ In_Progress, Resolved, Cancelled
- Resolved $\rightarrow$ Closed, In_Progress, Reopened
- Reopened $\rightarrow$ In_Progress, Resolved, Cancelled
- Closed / Cancelled: Terminal states (no further transitions permitted).

Answer Part 3 — Test DD and Traceability (10 pts)

Source File: <docs/lab-03/tests.md>

TokTickIT Lab 3 achieved a 100% automated test pass rate across all three testing tiers with zero skipped tests and zero regressions.

3.1 Traceability Matrix Summary

Every Acceptance Criterion (AC-01 to AC-18) and Business Rule (BR-01 to BR-18) is validated by dedicated automated test suites.

AC / BR	Description	Verifying Automated Test Suite	Status
AC-01 / BR-01	Valid login returns JWT; Inactive accounts blocked	server/tests/lab-03/auth.api.test.ts	PASS
AC-02 / BR-02	Mandatory password change on first login	client/tests/lab-03/ChangePassword.test.tsx / e2e/authentication.spec.ts	PASS
AC-03 / BR-03	Authenticated requester ownership protection	server/tests/lab-02/my-tickets.test.ts & ticket-detail.test.ts	PASS
AC-04 / BR-04	Internal notes 403 Forbidden to Requesters (Zero leakage)	server/tests/lab-03/comments-notes.api.test.ts	PASS
AC-05 / BR-05	Requester resolution indication vs. Staff formal resolution	server/tests/lab-03/staff-ticket-detail.api.test.ts	PASS
AC-06-08	Staff queue search, multi-filters, sorting, pagination	server/tests/lab-03/staff-queue.api.test.ts / StaffTicketQueue.test.tsx	PASS
AC-09-13	Ticket claim, reassign, IT priority, status matrix	server/tests/lab-03/staff-ticket-detail.api.test.ts / StaffTicketDetail.test.tsx	PASS
AC-16-18 / BR-13-15	Admin user management & safety invariants	server/tests/lab-03/users-admin.api.test.ts / UserManagement.test.tsx	PASS
E2E-01-12	Complete multi-role user journeys across 3 viewports	e2e/lab-03/*.spec.ts (Playwright)	PASS

3.2 Automated Test Execution Evidence

1. Server Vitest Test Suite (152/152 Passed):

```

✓ tests/lab-03/authorization.api.test.ts (10 tests) 74ms
✓ tests/lab-01/health.test.ts (1 test) 22ms
✓ tests/lab-03/staff-queue.api.test.ts (14 tests) 83ms
✓ tests/lab-03/staff-ticket-detail.api.test.ts (29 tests) 126ms
✓ tests/lab-03/comments-notes.api.test.ts (28 tests) 146ms
✓ tests/lab-01/categories.test.ts (1 test) 353ms
✓ tests/lab-02/related-systems.test.ts (1 test) 354ms
✓ tests/lab-02/requesters.test.ts (1 test) 356ms
✓ tests/lab-03/users-admin.api.test.ts (20 tests) 354ms
✓ tests/lab-02/create-ticket.test.ts (7 tests) 497ms
✓ tests/lab-02/ticket-detail.test.ts (4 tests) 522ms
✓ tests/lab-02/my-tickets.test.ts (8 tests) 577ms
✓ tests/lab-02/attachments.test.ts (11 tests) 660ms
✓ tests/lab-03/auth.api.test.ts (17 tests) 1395ms

```

```

Test Files 14 passed (14)
Tests     152 passed (152)
Duration  2.99s

```

2. Client Vitest Test Suite (68/68 Passed):

```
✓ tests/lab-02/RequesterSelect.test.tsx (2 tests) 116ms
✓ tests/lab-02/CreateTicket.test.tsx (4 tests) 144ms
✓ tests/lab-02/MyTickets.test.tsx (5 tests) 293ms
✓ tests/lab-02/TicketDetail.test.tsx (6 tests) 345ms
✓ tests/lab-02/AppShell.test.tsx (2 tests) 551ms
✓ tests/lab-01/App.test.tsx (3 tests) 525ms
✓ tests/lab-03/Login.test.tsx (9 tests) 784ms
✓ tests/lab-03/StaffTicketQueue.test.tsx (8 tests) 847ms
✓ tests/lab-03/StaffTicketDetail.test.tsx (11 tests) 920ms
✓ tests/lab-03/UserManagement.test.tsx (8 tests) 951ms
✓ tests/lab-03/ChangePassword.test.tsx (10 tests) 1743ms
```

```
Test Files 11 passed (11)
Tests      68 passed (68)
Duration   5.71s
```

3. Playwright Multi-Viewport E2E Test Suite (12/12 Passed):

```
ok 1 [chromium] > e2e/lab-03/authentication.spec.ts > TC-E2E-AUTH-01: valid login for Requester, Staff,
ok 2 [chromium] > e2e/lab-03/authentication.spec.ts > TC-E2E-AUTH-02: mandatory password change flow on
ok 3 [chromium] > e2e/lab-03/authentication.spec.ts > TC-E2E-AUTH-03: invalid credentials display clear
ok 4 [chromium] > e2e/lab-03/authentication.spec.ts > TC-E2E-AUTH-04: inactive account is blocked with
ok 5 [chromium] > e2e/lab-03/authentication.spec.ts > TC-E2E-AUTH-05: multi-viewport responsiveness and
ok 6 [chromium] > e2e/lab-03/staff-ticket-flow.spec.ts > TC-E2E-STAFF-01: full ticket lifecycle – creat
ok 7 [chromium] > e2e/lab-03/staff-ticket-flow.spec.ts > TC-E2E-STAFF-02: multi-viewport responsiveness
ok 8 [chromium] > e2e/lab-03/user-administration.spec.ts > TC-E2E-ADMIN-01: Admin user directory viewin
ok 9 [chromium] > e2e/lab-03/user-administration.spec.ts > TC-E2E-ADMIN-02: User creation with temporar
ok 10 [chromium] > e2e/lab-03/user-administration.spec.ts > TC-E2E-ADMIN-03: Safety invariants enforceme
ok 11 [chromium] > e2e/lab-03/user-administration.spec.ts > TC-E2E-ADMIN-04: Password reset provisioning
ok 12 [chromium] > e2e/lab-03/user-administration.spec.ts > TC-E2E-ADMIN-05: Multi-viewport responsivene

12 passed (15.7s)
```

Answer Part 4 — AI Use with Reflection (5 pts)

Source File: [docs/lab-03/ai-use.md](#)

LLM / AI System: Google Antigravity IDE (Gemini 2.5 Pro Agentic Pair-Programming System)

4.1 Prompts & Human Guidance

During Sprint 3, 9 key prompts were formulated (detailed in [docs/lab-03/ai-use.md](#)) to guide specification generation, database migration, authentication routes, staff operations, confidential comments/notes, administrator user management, and Playwright E2E suites.

4.2 Reflection on Engineering Collaboration

Working with the agentic AI provided high velocity during TDD boilerplate creation and comprehensive test authoring. However, critical engineering oversights required human intervention:

1. **Concurrency Protection:** The AI generated a sequential check for the remaining active Administrator count before deactivating. Human peer review correctly identified that concurrent requests could bypass the check. We intervened and wrapped the query in a transactional block with serializable isolation.

2. **Deterministic UI Assertions:** In Playwright tests, multiple elements containing the user name caused ambiguity in strict mode. Introducing semantic `.user-name` and `.user-role-badge` CSS classes eliminated locator collisions.
3. **Decoupling System Counts from UI Filter State:** In user management, calculating the active admin count from filtered search results falsely triggered the "sole active admin" protection. We corrected the state model to maintain global system counts independently of active search terms.

Answer Part 5 — Working Login and Password Change UI (5 pts)

The authentication system implements secure JWT-based access control with clean Zen Green styling, comprehensive input validation, busy states, and safe error feedback.

5.1 Login Screen & Validation

The Login screen allows users to authenticate with their email and password. A helper panel provides 1-click test credentials for Requester, IT Staff, and Administrator roles during demonstration.

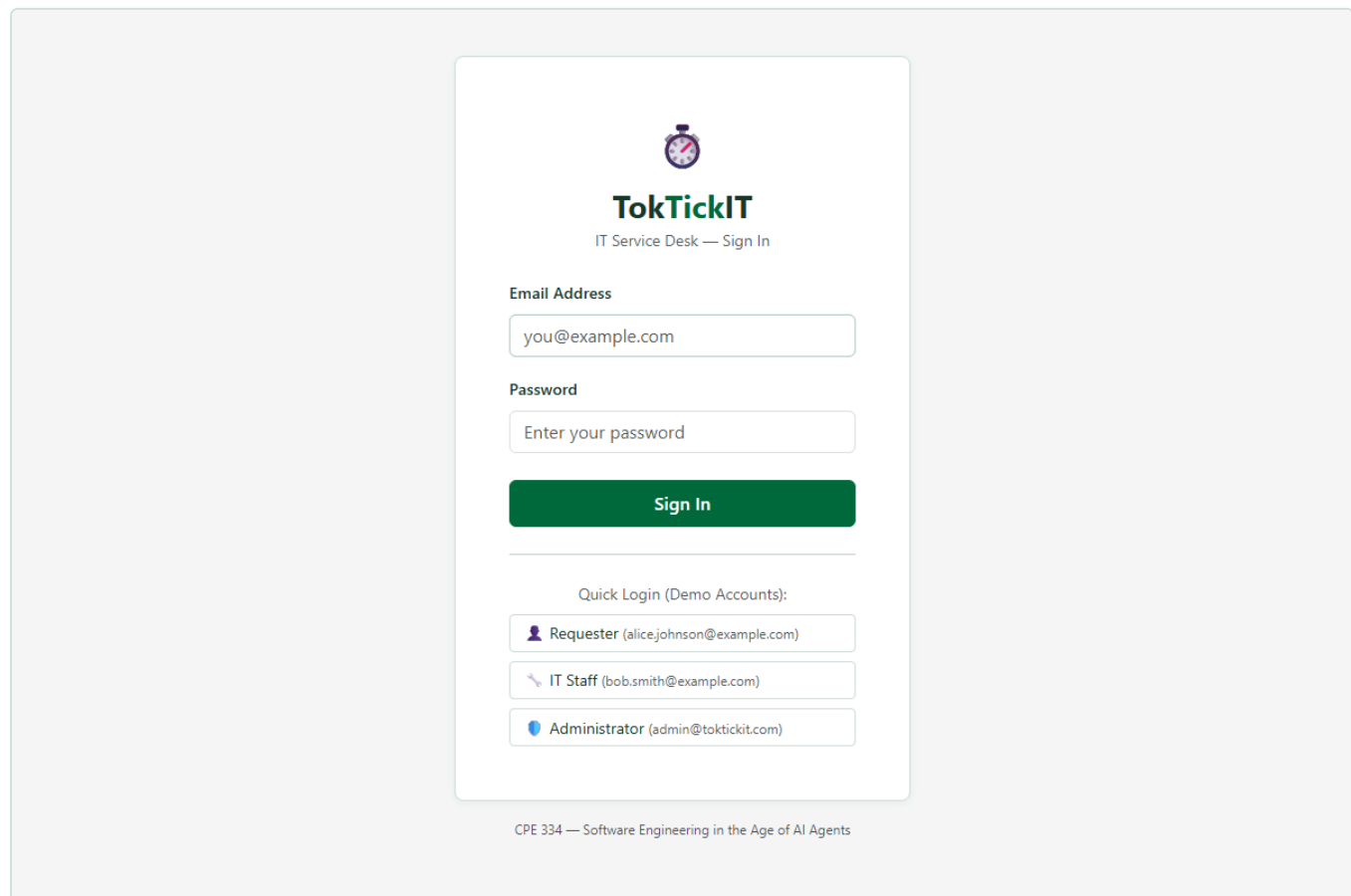
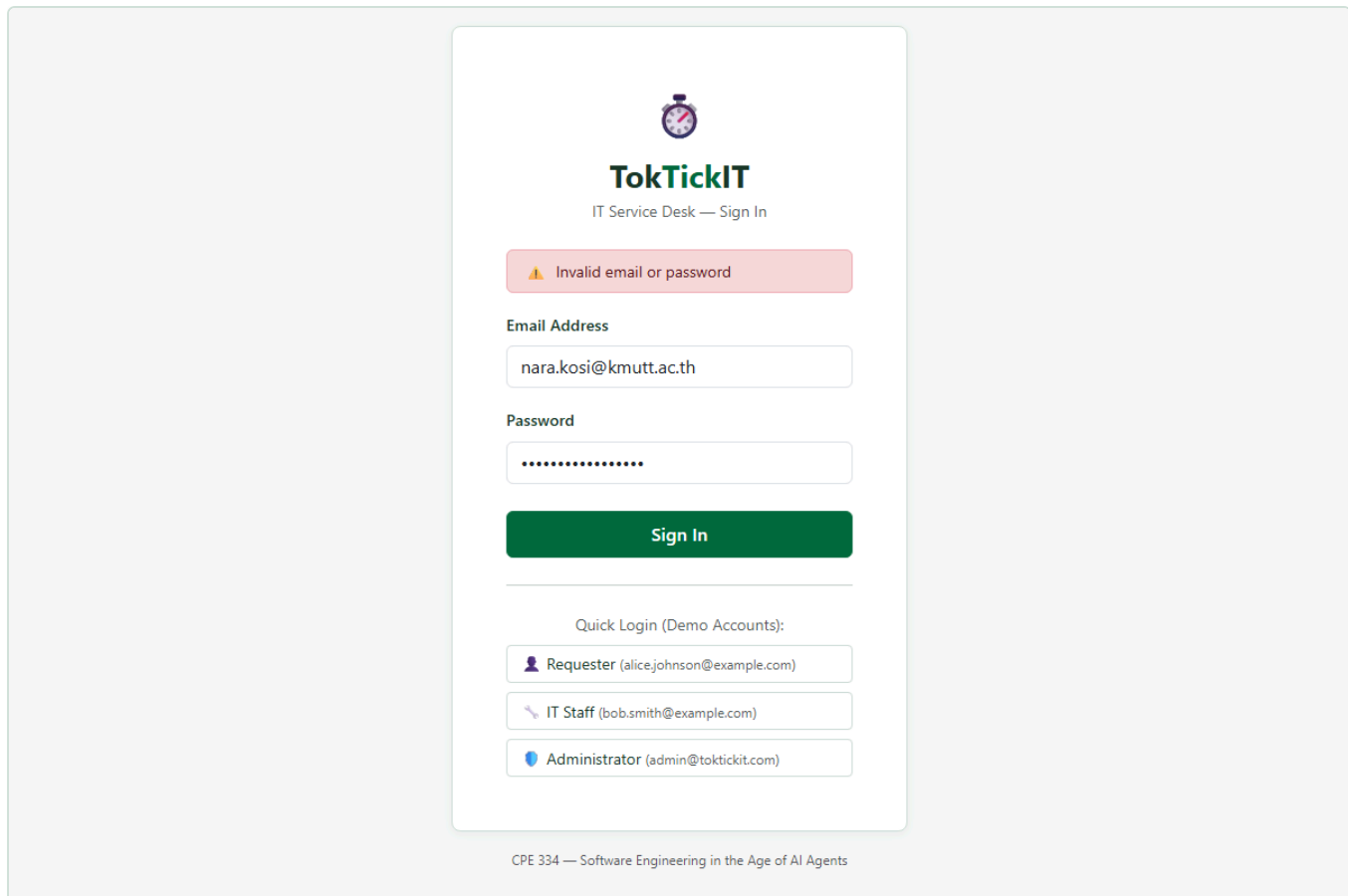
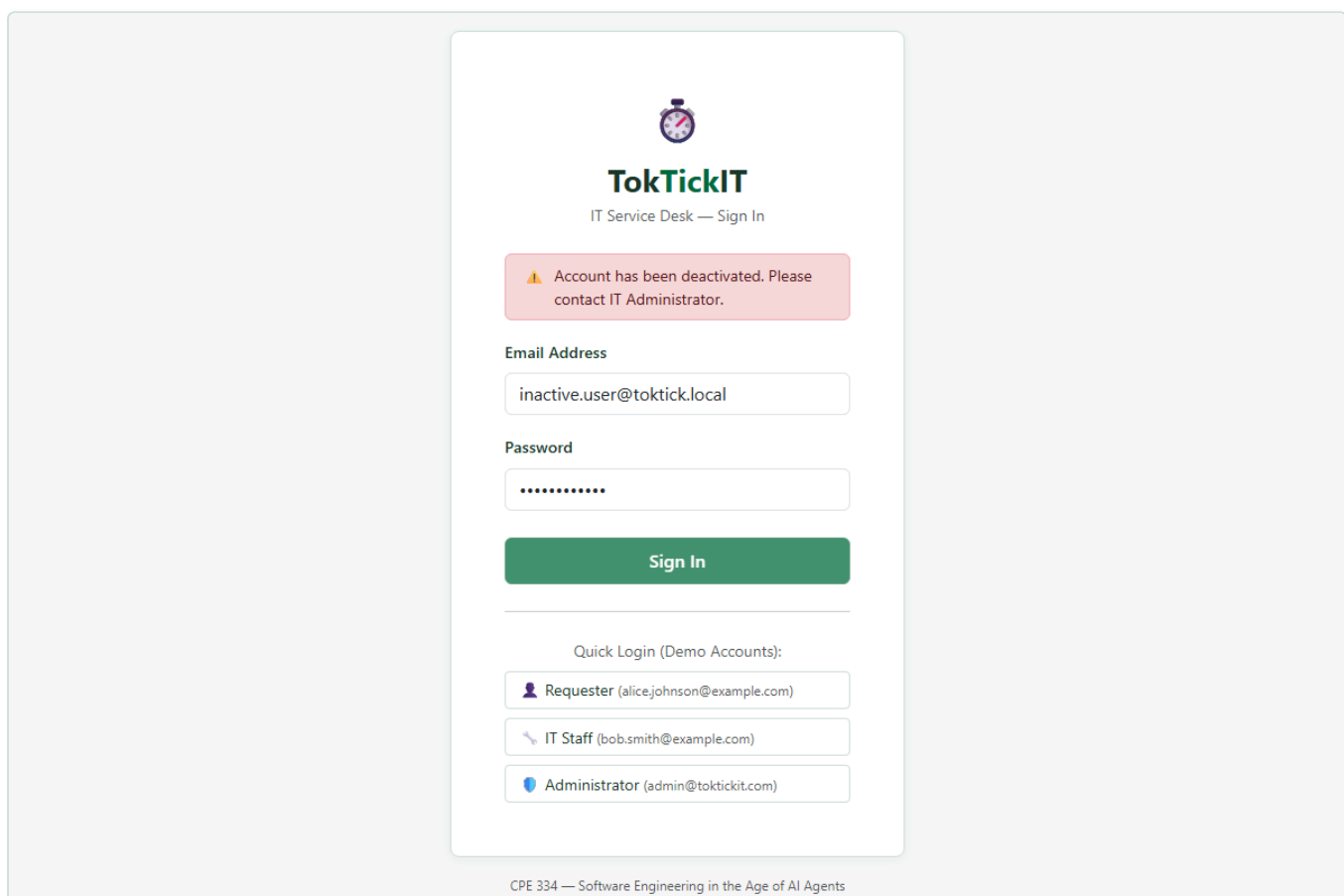


Figure 5.1: Login screen on Desktop (1200px) showing Zen Green card, form fields, and quick-login helpers.



The screenshot shows the TokTickIT login interface. At the top is a clock icon and the text "TokTickIT IT Service Desk — Sign In". Below this is a red error message box with a warning icon and the text "Invalid email or password". The "Email Address" field contains "nara.kosi@kmutt.ac.th" and the "Password" field is masked with dots. A green "Sign In" button is below the fields. Underneath is a "Quick Login (Demo Accounts):" section with three buttons: "Requester (alice.johnson@example.com)", "IT Staff (bob.smith@example.com)", and "Administrator (admin@toktickit.com)". At the bottom, it says "CPE 334 — Software Engineering in the Age of AI Agents".

Figure 5.2: Invalid credentials alert — displays generic safe error without revealing account existence.

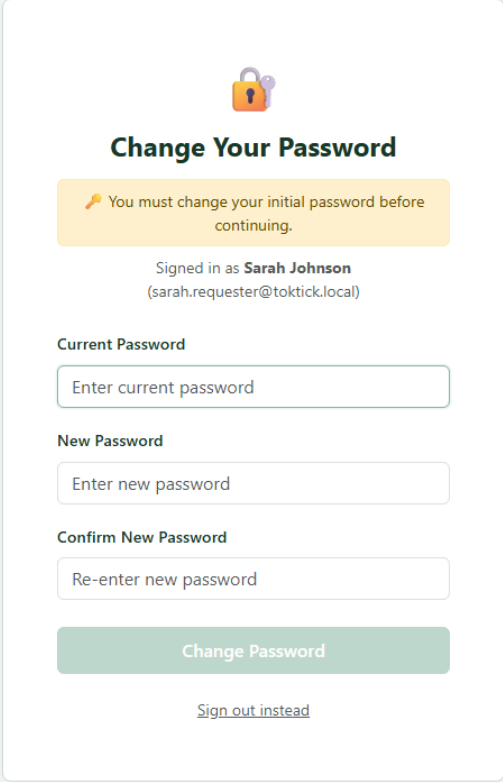


The screenshot shows the TokTickIT login interface. At the top is a clock icon and the text "TokTickIT IT Service Desk — Sign In". Below this is a red error message box with a warning icon and the text "Account has been deactivated. Please contact IT Administrator.". The "Email Address" field contains "inactive.user@toktick.local" and the "Password" field is masked with dots. A green "Sign In" button is below the fields. Underneath is a "Quick Login (Demo Accounts):" section with three buttons: "Requester (alice.johnson@example.com)", "IT Staff (bob.smith@example.com)", and "Administrator (admin@toktickit.com)". At the bottom, it says "CPE 334 — Software Engineering in the Age of AI Agents".

Figure 5.3: Inactive account blocked (BR-01) — displays clear deactivation notice while rejecting authentication.

5.2 Mandatory First-Login Password Change (BR-02)

When a user with `mustChangePassword: true` logs in, the application intercepts access and directs the user to the mandatory Change Password screen. The screen features a live visual complexity checklist (8+ characters, uppercase, lowercase, number, special symbol) and confirmation match validation.





Change Your Password

 You must change your initial password before continuing.

Signed in as **Sarah Johnson**
(sarah.requester@toktick.local)

Current Password

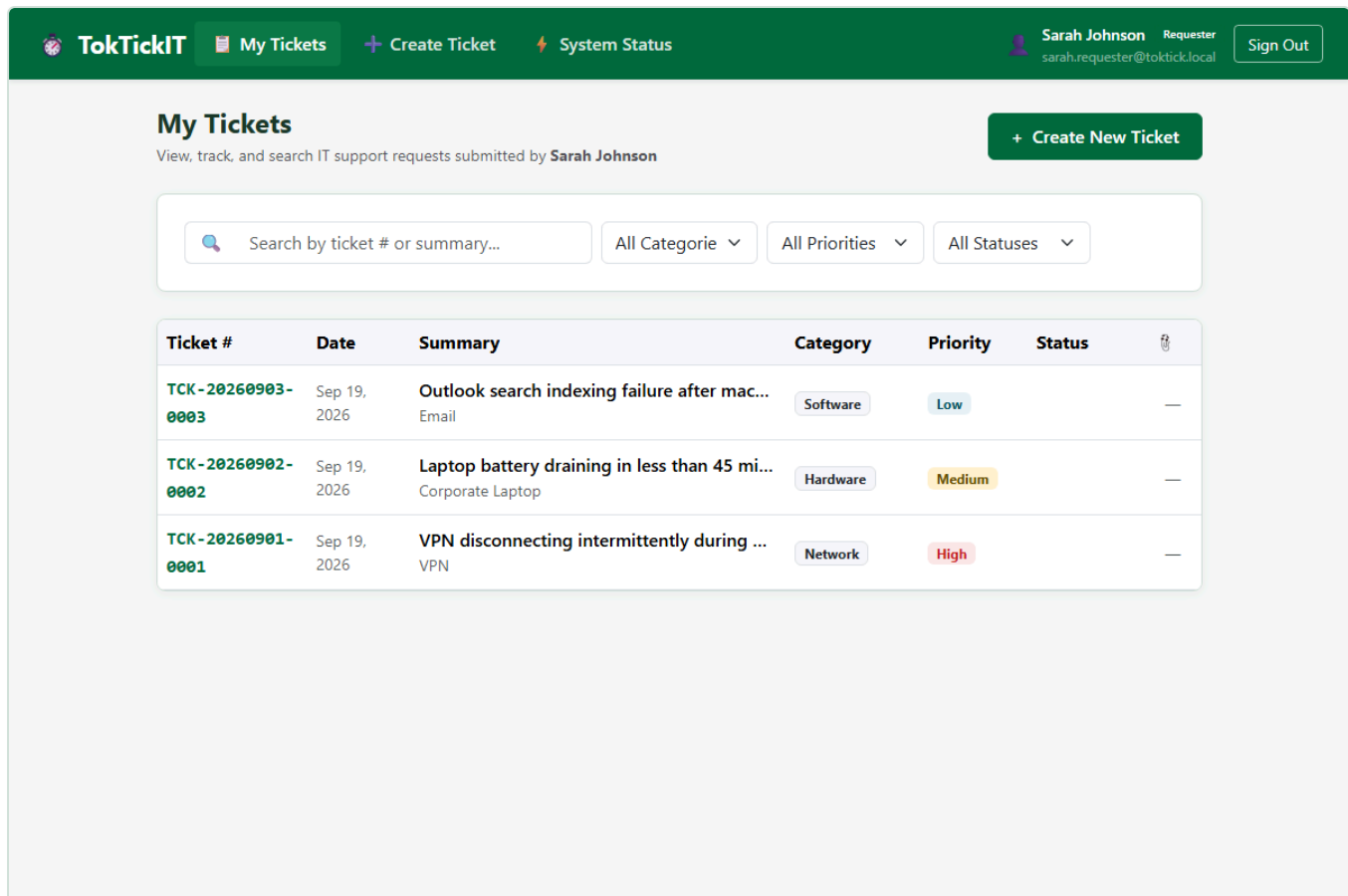
New Password

Confirm New Password

Change Password

[Sign out instead](#)

Figure 5.4: Mandatory password change screen showing dynamic complexity checklist and current password verification.



TokTickIT My Tickets Create Ticket System Status Sarah Johnson Requester sarah.requester@toktick.local Sign Out

My Tickets

View, track, and search IT support requests submitted by Sarah Johnson

+ Create New Ticket

Search by ticket # or summary... All Categories All Priorities All Statuses

Ticket #	Date	Summary	Category	Priority	Status	
TCK-20260903-0003	Sep 19, 2026	Outlook search indexing failure after mac... Email	Software	Low		—
TCK-20260902-0002	Sep 19, 2026	Laptop battery draining in less than 45 mi... Corporate Laptop	Hardware	Medium		—
TCK-20260901-0001	Sep 19, 2026	VPN disconnecting intermittently during ... VPN	Network	High		—

Figure 5.5: Successful password change redirect — clears `mustChangePassword` flag and routes user to their role dashboard.

Answer Part 6 — Working IT Staff Ticket Queue UI (5 pts)

The shared IT Staff Ticket Queue (`/staff/queue`) empowers staff to locate, prioritize, and manage workload efficiently.

6.1 Queue Data, Filters, Sorting & Pagination

The queue displays realistic ticket data with Ticket Number, Created Date, Summary, Category, Requested Priority, IT Priority, Status, and Ticket Owner.

 **TokTickIT**

[Ticket Queue](#) [My Tickets](#) [Create Ticket](#) [System Status](#)

Charlie Staff IT Staff

charlie.staff@toktick.local

Sign Out

 **IT Support Queue**

Shared service desk queue for IT Support Staff and Administrators

260 Total Tickets [Refresh](#)

Total Queue

260

New

253

In Progress

3

Pending Requester

0

Resolved

0

Search

Status

Priority

Category

Assignee

Clear

Search ticket #, title, summary...

All Statuses

All

All Categories

All Tickets

Clear

ID / TICKET NO	TITLE	REQUESTER	CATEGORY	STATUS	PRIORITY	ASSIGNEE	CREATED AT	ACTION
TKT-2026-000258	[E2E] VPN connectivity drop during ... 1 1	Alice Johnson alice.johnson@example....	Network	Closed	IT: Critical	Charlie Staff	Sep 20, 2026	Open Details
TKT-2026-000257	[E2E] VPN connectivity drop during ...	Alice Johnson alice.johnson@example....	Network	New	High	Unassigned	Sep 20, 2026	Open Details
TKT-2026-000256	Max 5 Attachments Ticket 5	Jennifer Anderson jennifer.anderson@exa...	Account and Access	New	Low	Unassigned	Sep 20, 2026	Open Details
TKT-								

Figure 6.1: Shared IT Staff Ticket Queue on Desktop (1200px) showing summary metric cards, filter controls, and paginated table.

**TokTickIT**

 Ticket Queue

 My Tickets

 Create Ticket

 System Status

**Charlie Staff** IT Staff

charlie.staff@toktickit.local

Sign Out

**IT Support Queue**

260 Total Tickets

 Refresh

Shared service desk queue for IT Support Staff and Administrators

Total Queue
260

New
253

In Progress
3

Pending Requester
0

Resolved
0

Search

 Search ticket #, title, summary...

Status

Priority

Category

Assignee

All Statuses

All

All Categories

All Tickets

Clear

ID / TICKET NO	TITLE	REQUESTER	CATEGORY	STATUS
TKT-2026-000258	[E2E] VPN connectivity drop during ...  1  1	Alice Johnson alice.johnson@example....	Network	Close
TKT-2026-000257	[E2E] VPN connectivity drop during ...	Alice Johnson alice.johnson@example....	Network	New
TKT-2026-000256	Max 5 Attachments Ticket  5	Jennifer Anderson jennifer.anderson@exa...	Account and Access	New
TKT-2026-000255	Second test ticket for number check	Jennifer Anderson jennifer.anderson@exa...	Hardware	New

Figure 6.2: Staff Ticket Queue on Tablet (800px) maintaining clear column alignment and full readability.

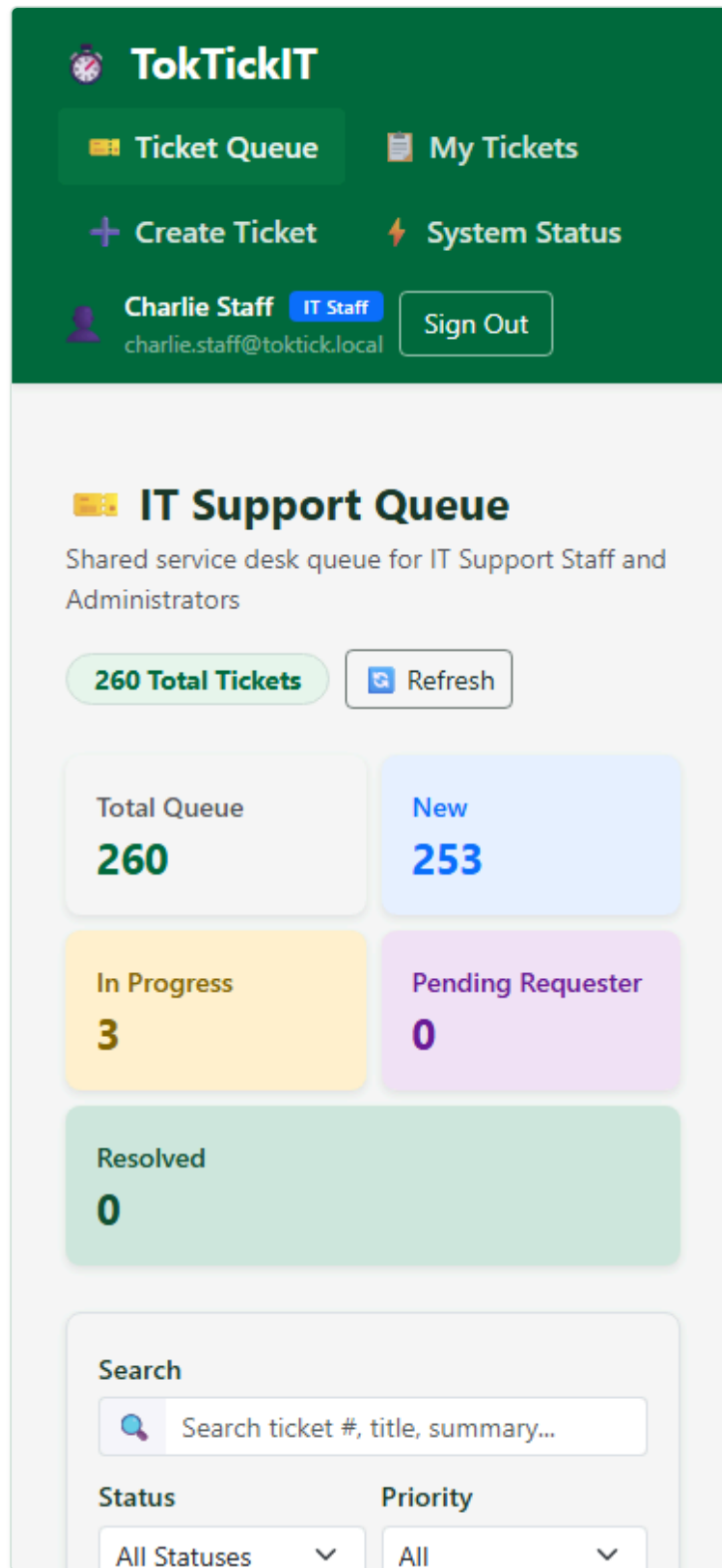


Figure 6.3: Staff Ticket Queue on Mobile (375px) automatically transitioning to responsive stacked ticket cards with zero horizontal overflow.

Answer Part 7 — Working IT Staff Ticket Detail UI (10 pts)

The IT Staff Ticket Detail screen extends the ticket view with full operational capabilities: ticket ownership assignment, IT Priority override, status transitions, public communication, and confidential internal notes.

7.1 Operational Controls & IT Priority Override

Staff can claim unassigned tickets or reassign ownership to any active IT Staff member. Staff can also adjust **IT Priority** (e.g. Critical, High, Medium, Low) without mutating the original user-submitted **Requested Priority**.

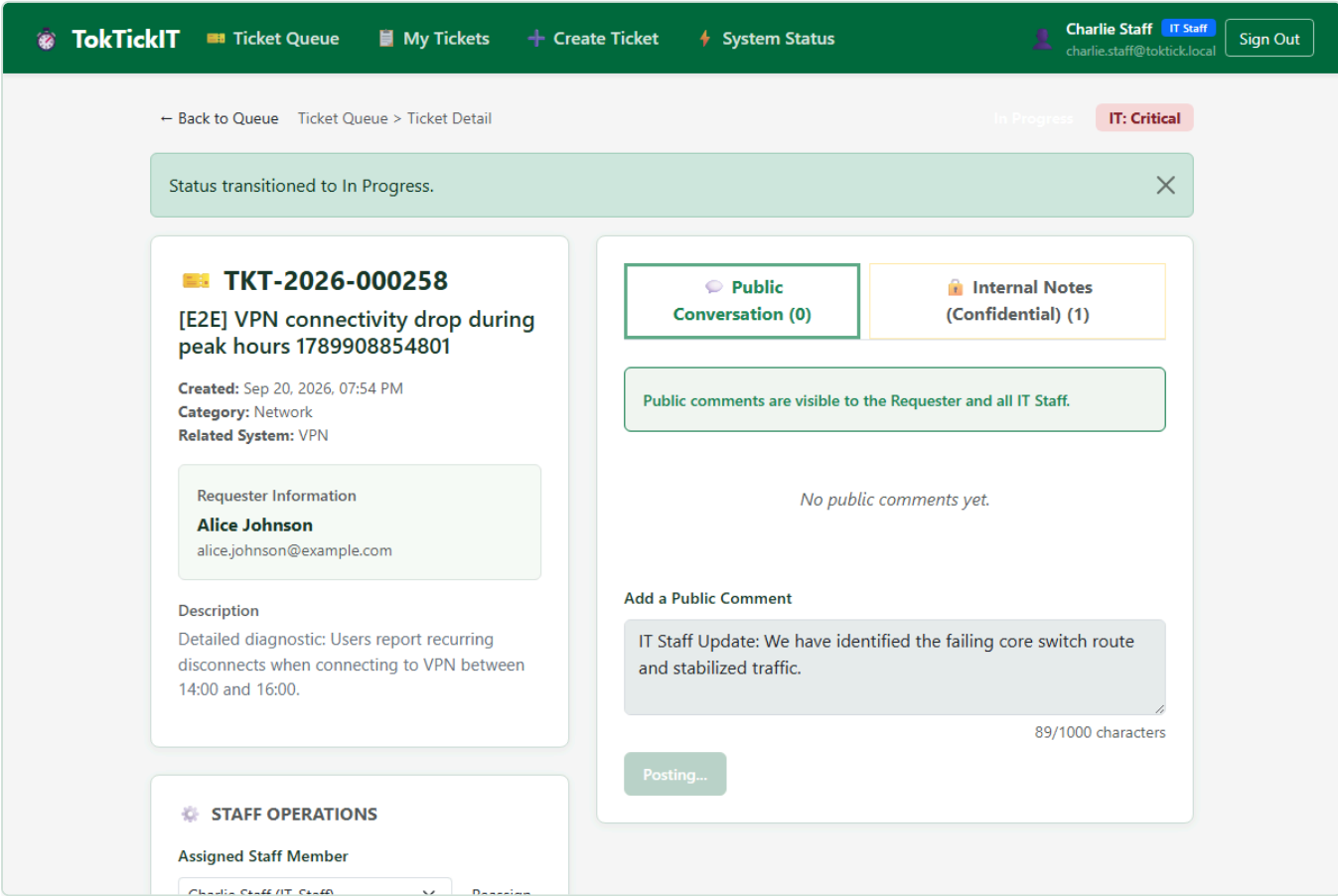


Figure 7.1: Staff Ticket Detail on Desktop showing ownership controls, IT Priority override selector, and permitted status transitions.

7.2 Public Comments vs. Confidential Internal Notes (BR-04)

The conversation area features dual tabs:

- **Public Conversation:** Shared timeline visible to Requester, IT Staff, and Administrator.
- **Internal Notes (Confidential):** Operational notes restricted strictly to IT Staff and Administrator. The tab is marked with an amber warning banner (

Confidential — Visible only to IT Staff and Administrators

).

← Back to Queue Ticket Queue > Ticket Detail In Progress **IT: Critical**

Status transitioned to In Progress. ✕

TKT-2026-000258
[E2E] VPN connectivity drop during peak hours 1789908854801

Created: Sep 20, 2026, 07:54 PM
Category: Network
Related System: VPN

Requester Information
Alice Johnson
alice.johnson@example.com

Description
Detailed diagnostic: Users report recurring disconnects when connecting to VPN between 14:00 and 16:00.

STAFF OPERATIONS

Assigned Staff Member
Charlie Staff (IT Staff)

Public Conversation (1)

Internal Notes (Confidential) (1)

Staff Eyes Only
Internal Notes are confidential and visible strictly to IT Staff and Administrators. Requesters cannot see these notes.

Charlie Staff IT Staff Sep 20, 2026, 07:54 PM
Staff Internal Note: Diagnosed flapping gateway interface on core switch SW-02. Scheduled reboot.

Add a Confidential Internal Note

Record private technical notes, vendor tickets, or diagnosis details...

0/1000 characters

Save Internal Note

Figure 7.2: Confidential Internal Notes tab with distinct amber styling and operational discussion.

← Back to My Tickets

Problem Appears Resolved High Priority In Progress

TKT-2026-000258 Submitted on Sep 20, 2026, 07:54 PM

[E2E] VPN connectivity drop during peak hours 1789908854801

Category	Related System	Requester	Contact Email
Network	VPN	Alice Johnson	alice.johnson@example.com

Description

Detailed diagnostic: Users report recurring disconnects when connecting to VPN between 14:00 and 16:00.

Attachments
(0 / 5 active attachments)

+ Add Attachment

No attachments uploaded for this ticket.

Public Conversation (1) Visible to Support Staff

Figure 7.3: Requester Ticket Detail view — shows Public Conversation and "Problem Appears Resolved" button with zero leakage of internal notes.

7.3 Terminal Closure & Status Invariants

Once a ticket reaches terminal state (`Closed` or `Cancelled`), operational controls are frozen to preserve audit integrity.

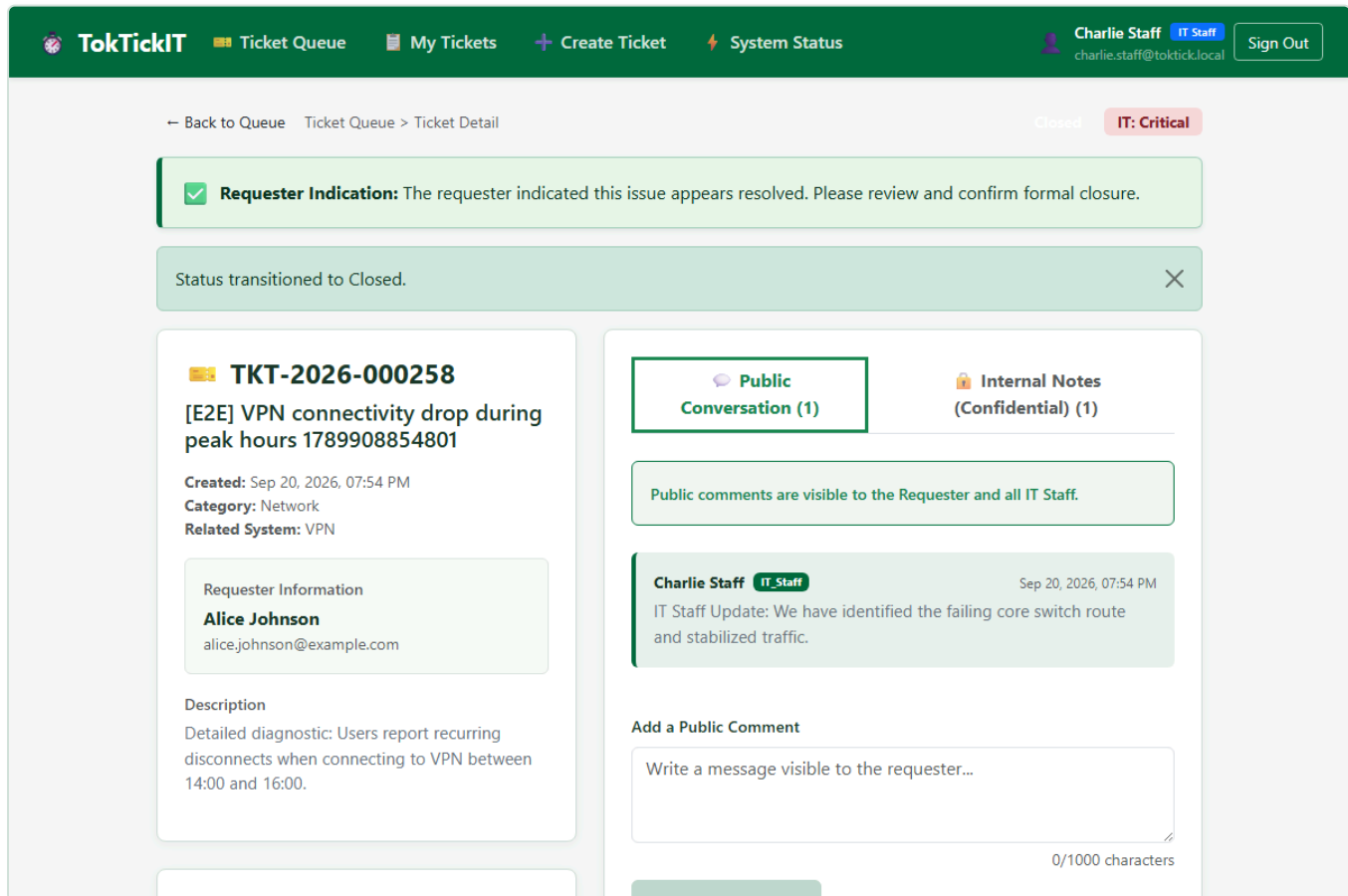


Figure 7.4: Formally Closed ticket in terminal state with read-only badges and disabled transition actions.

Answer Part 8 — Working Administrator User Management UI (5 pts)

The Administrator interface provides a minimalist, robust user administration directory supporting account creation, editing, password resets, and safety invariant enforcement.

8.1 User Directory, Search & Filtering

The directory lists all system accounts with Name, Email, Role badge, Active status pill, and Edit action. Administrators can filter by role and perform debounced text search across names and emails.

User Management
Ticket Queue
My Tickets
Create Ticket
System Status

Alice Admin Admin
Sign Out

User Management

Manage user accounts, roles, access permissions, and temporary password provisioning.

All Roles

Name	Email	Role	Status	Password Status	Created Date	Actions
Jennifer Anderson	jennifer.anderson@example.com	Requester	Active	Must Change	Sep 4, 2026	Edit Reset Password
Michael Brown	michael.brown@example.com	Requester	Active	Must Change	Sep 4, 2026	Edit Reset Password
Sarah Johnson	sarah.johnson@example.com	Requester	Active	Must Change	Sep 4, 2026	Edit Reset Password
David Lee	david.lee@example.com	Requester	Active	Must Change	Sep 4, 2026	Edit Reset Password
Inactive Tester	inactive.test@example.com	Requester	Inactive	Must Change	Sep 4, 2026	Edit Reset Password
Nara Kosiyaorn	nara.kosi@kmutt.ac.th	Requester	Active	Normal	Sep 5, 2026	Edit Reset Password
Sunny farmhouse	nara2012sun@gmail.com	Requester	Active	Normal	Sep 5, 2026	Edit Reset Password
Alice Admin You	admin@toktick.local	Administrator	Active	Normal	Sep 19, 2026	Edit Reset Password
Charlie Staff	charlie.staff@toktick.local	IT Staff	Active	Normal	Sep 19, 2026	Edit Reset Password

Figure 8.1: Administrator User Management directory on Desktop showing user list, role badges, and status pills.

User Management
Ticket Queue
My Tickets
Create Ticket
System Status

Alice Admin Admin
Sign Out

User Management

Manage user accounts, roles, access permissions, and temporary password provisioning.

Administrator

Name	Email	Role	Status	Password Status	Created Date	Actions
Alice Admin You	admin@toktick.local	Administrator	Active	Normal	Sep 19, 2026	Edit Reset Password
Admin User	admin@toktickit.com	Administrator	Active	Normal	Sep 20, 2026	Edit Reset Password

Figure 8.2: Live filtering by role (IT Staff) and text search.

8.2 Account Creation & Temporary Credential Provisioning

Administrators can create users with auto-generated temporary passwords. The creation modal displays the credentials with a 1-click clipboard copy button and informs that the user will be prompted to change their password upon initial login.

The screenshot displays the TokTickIT User Management interface. A modal titled "Add New User" is open, allowing an administrator to create a new user. The modal includes the following fields and options:

- Full Name ***: A text input field containing "E2E Staff Member 254".
- Email Address ***: A text input field containing "e2e.staff.1789908860254@toktick.local".
- Role ***: A dropdown menu with the selected role "IT Staff (Shared ticket queue, notes, ops)".
- Initial Password (optional)**: A text input field with the placeholder "Leave blank to auto-generate strong password". Below this field, a note states: "If left blank, a random compliant temporary password will be generated."
- Account Active immediately**: A toggle switch that is currently turned on.
- Buttons**: "Cancel" and "Create User" buttons at the bottom right of the modal.

The background interface shows the "User Management" section with a search bar and a table of existing users. The table has columns for Name, Email, Role, Status, and Actions.

Name	Email	Role	Status	Created Date	Actions
Jennifer Anderson	jennifer.anderson@e			Sep 4, 2026	Edit Reset Password
Michael Brown	michael.brown@exa			Sep 4, 2026	Edit Reset Password
Sarah Johnson	sarah.johnson@exa			Sep 4, 2026	Edit Reset Password
David Lee	david.lee@example			Sep 4, 2026	Edit Reset Password
Inactive Tester	inactive.test@exa			Sep 4, 2026	Edit Reset Password
Nara Kosiyporn	nara.kosi@kmutt.ac			Sep 5, 2026	Edit Reset Password
Sunny farmhouse	nara2012sun@gmail			Sep 5, 2026	Edit Reset Password
Alice Admin	admin@toktick.local	Administrator	Active	Sep 19, 2026	Edit Reset Password
Charlie Staff	charlie.staff@toktick.local	IT Staff	Active	Sep 19, 2026	Edit Reset Password

Figure 8.3: Add User modal form with role selection and input validation.

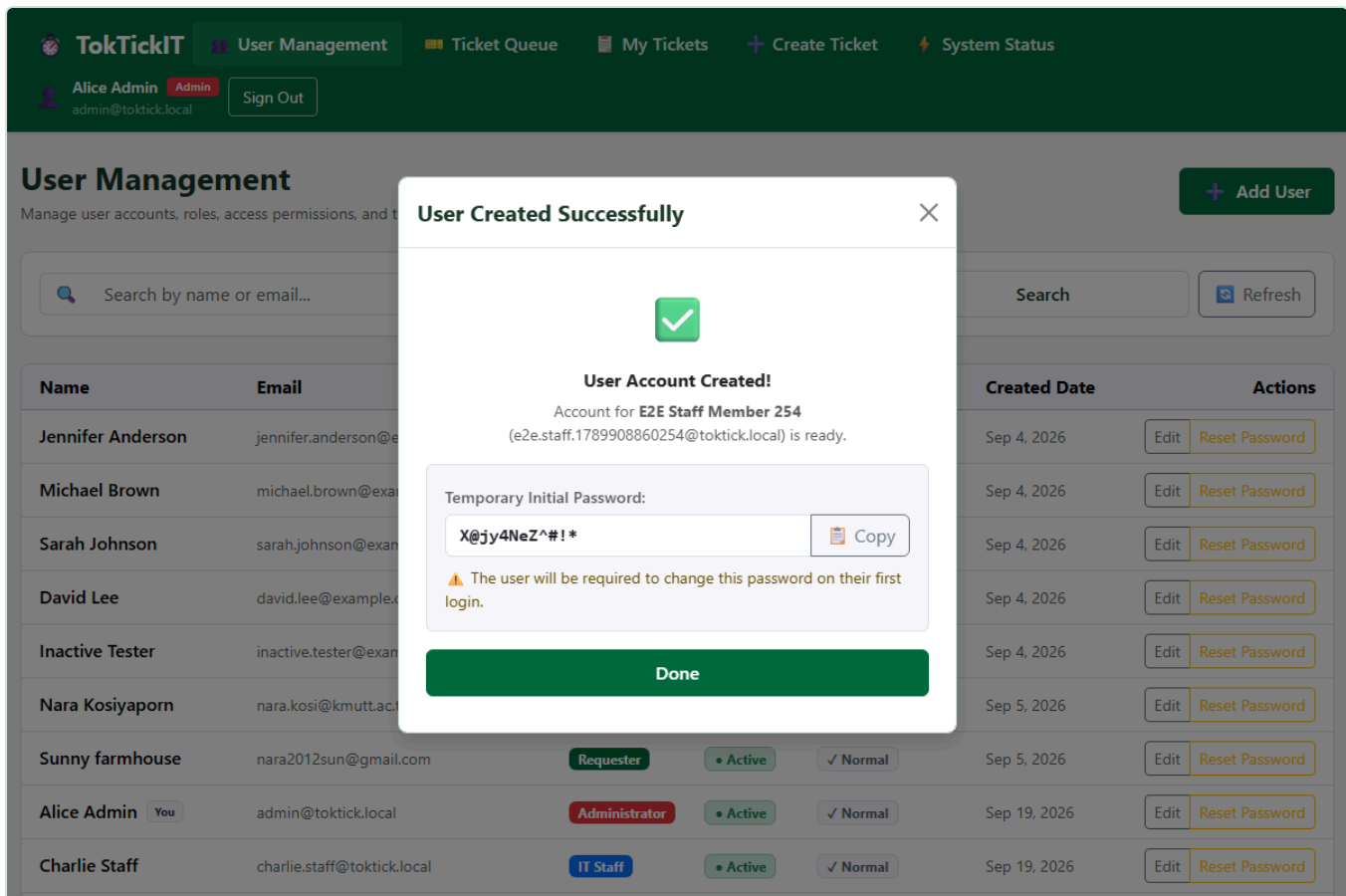


Figure 8.4: User creation success banner displaying auto-generated temporary password and copy helper.

8.3 Safety Invariant Protection (BR-13, BR-14, BR-15)

- **Self-Deactivation Prevention (BR-13):** When the currently logged-in administrator attempts to edit themselves, the Active switch is disabled with warning:
"You cannot deactivate your own account."
- **Sole Active Administrator Protection (BR-14):** When editing the last remaining active administrator, both role modification and deactivation are locked with warning:
"Cannot deactivate or change role of the last active Administrator."
- **Hard Deletion Forbidden (BR-15):** User deletion is rejected at the API layer with HTTP 405 Method Not Allowed; soft-deactivation must be used instead.

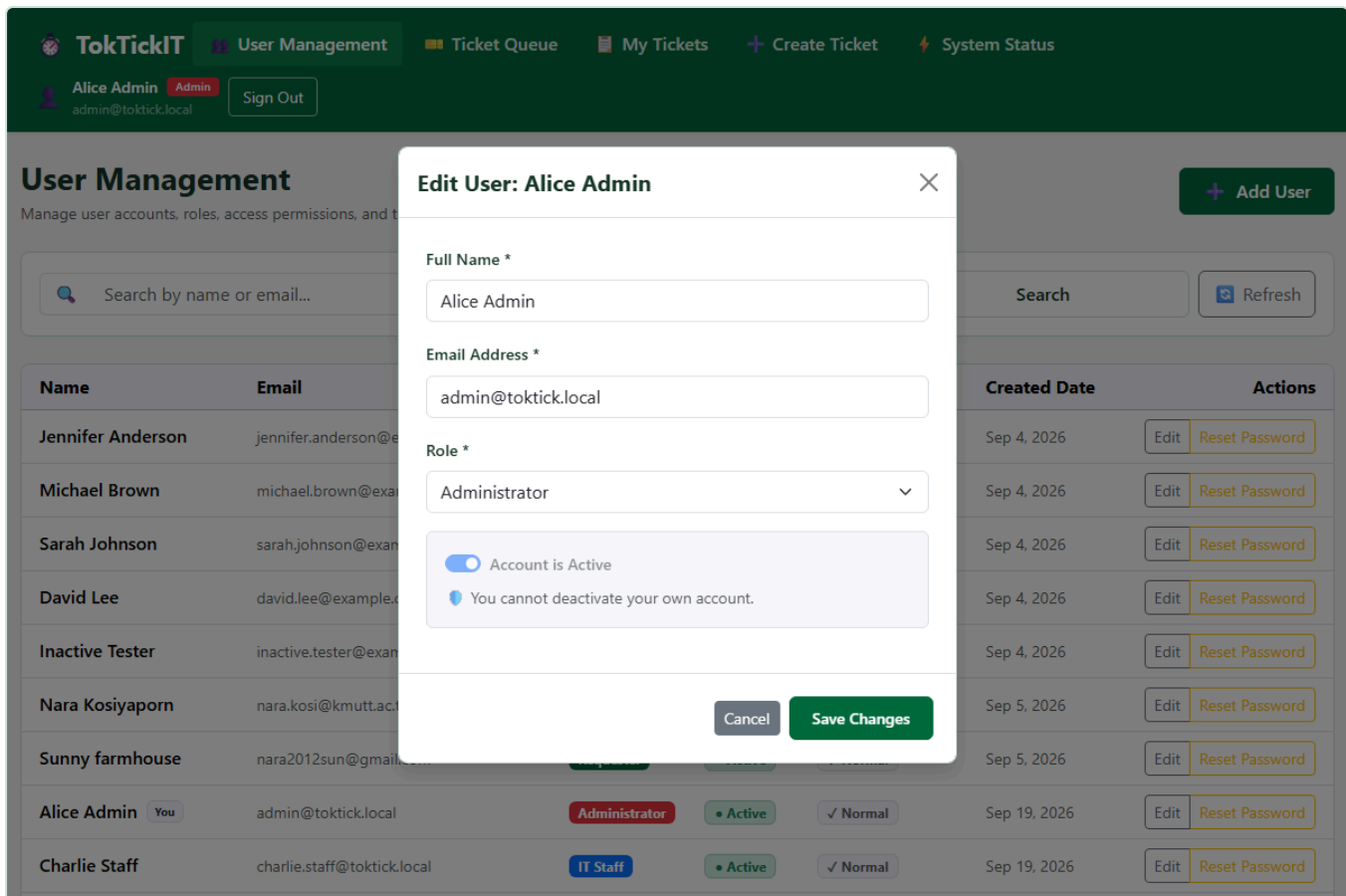


Figure 8.5: Safety invariant UX — Active switch disabled with explanatory warning when an administrator edits their own account.

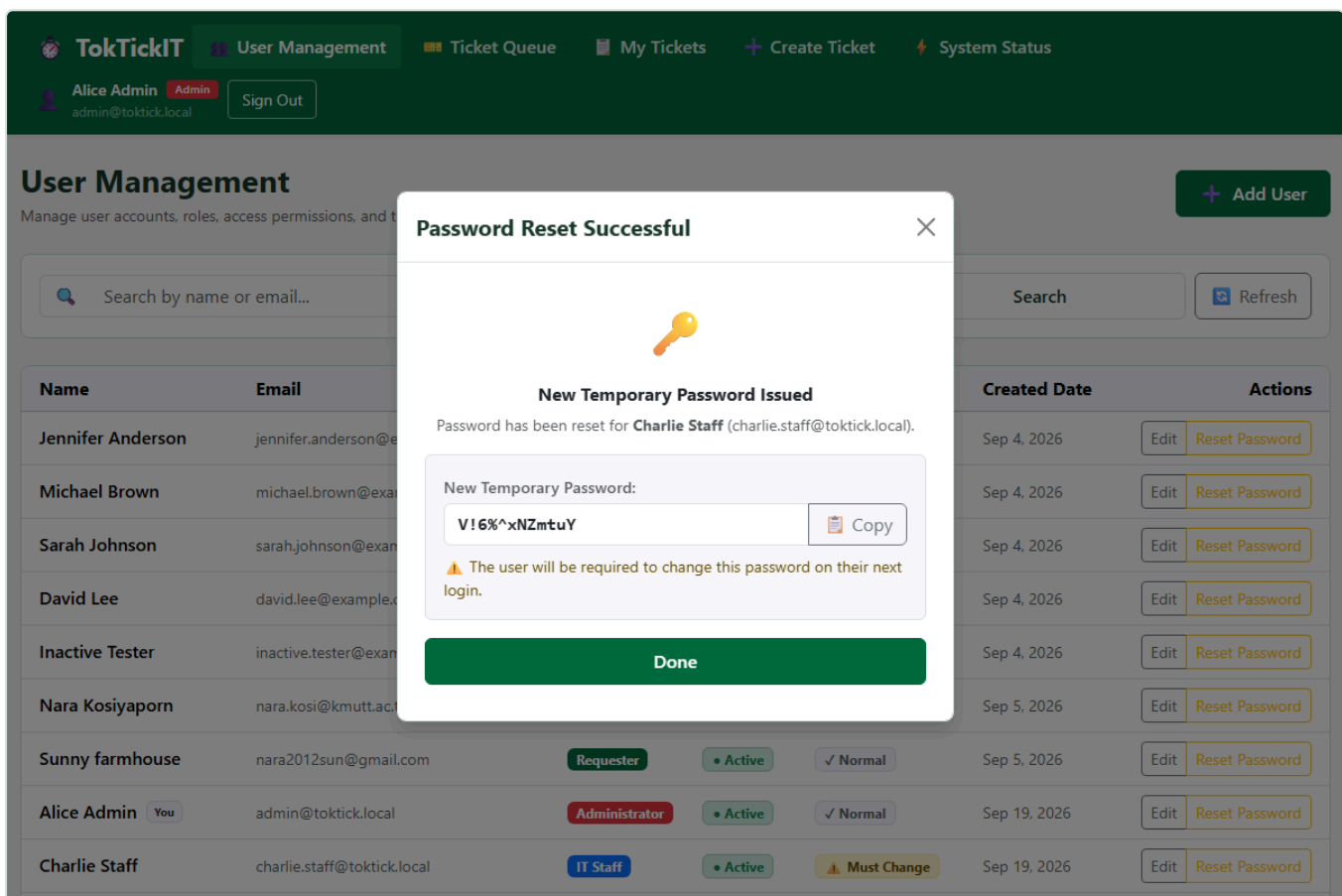


Figure 8.6: Administrative password reset modal generating new temporary credentials and resetting `mustChangePassword: true`.

Answer Part 9 — Zen Green UI and Responsive Evidence (5 pts)

Source File: docs/lab-03/ui-spec.md

All screens strictly follow the Zen Green design system tokens established in Lab 2 (--zen-primary: #006B3C , --zen-secondary: #0B7A46 , --zen-pale: #EAF6EF , --zen-dark: #1B3A2A).

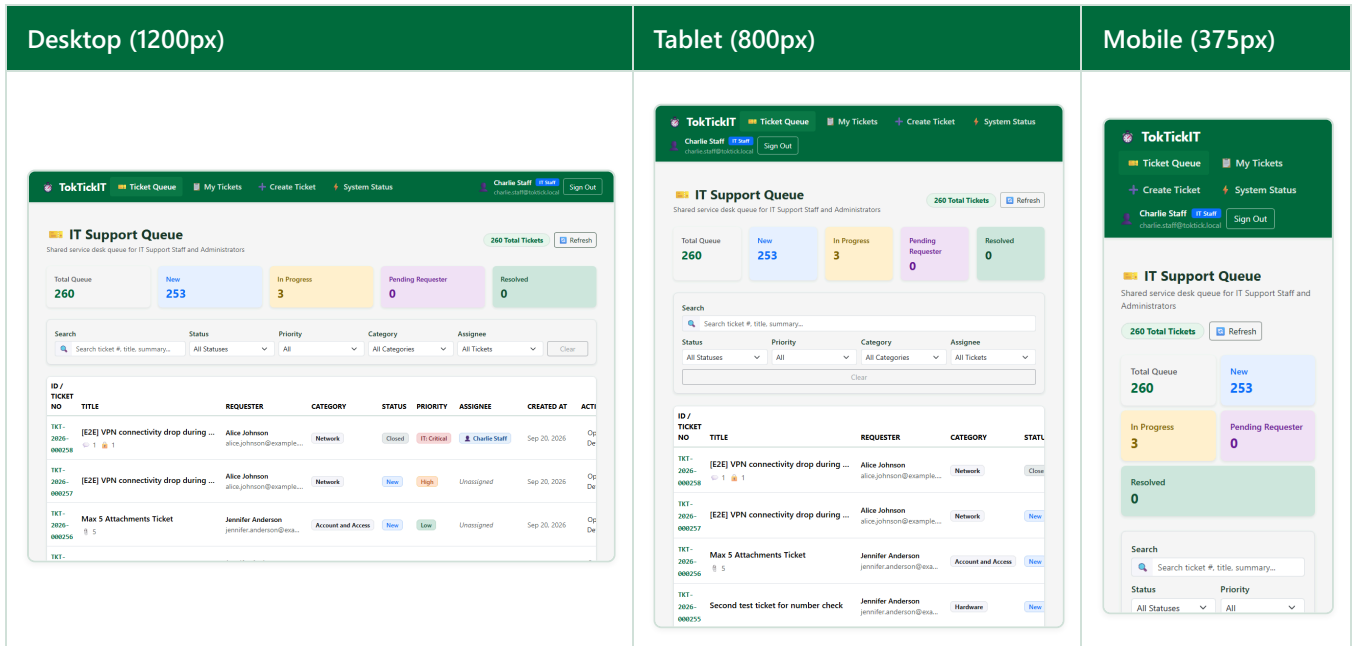
9.1 Multi-Viewport Verification

Every required screen was tested across Desktop (1200px), Tablet (800px), and Mobile (375px) viewports with Playwright automated assertions verifying zero horizontal overflow (document.documentElement.scrollWidth <= document.documentElement.clientWidth).

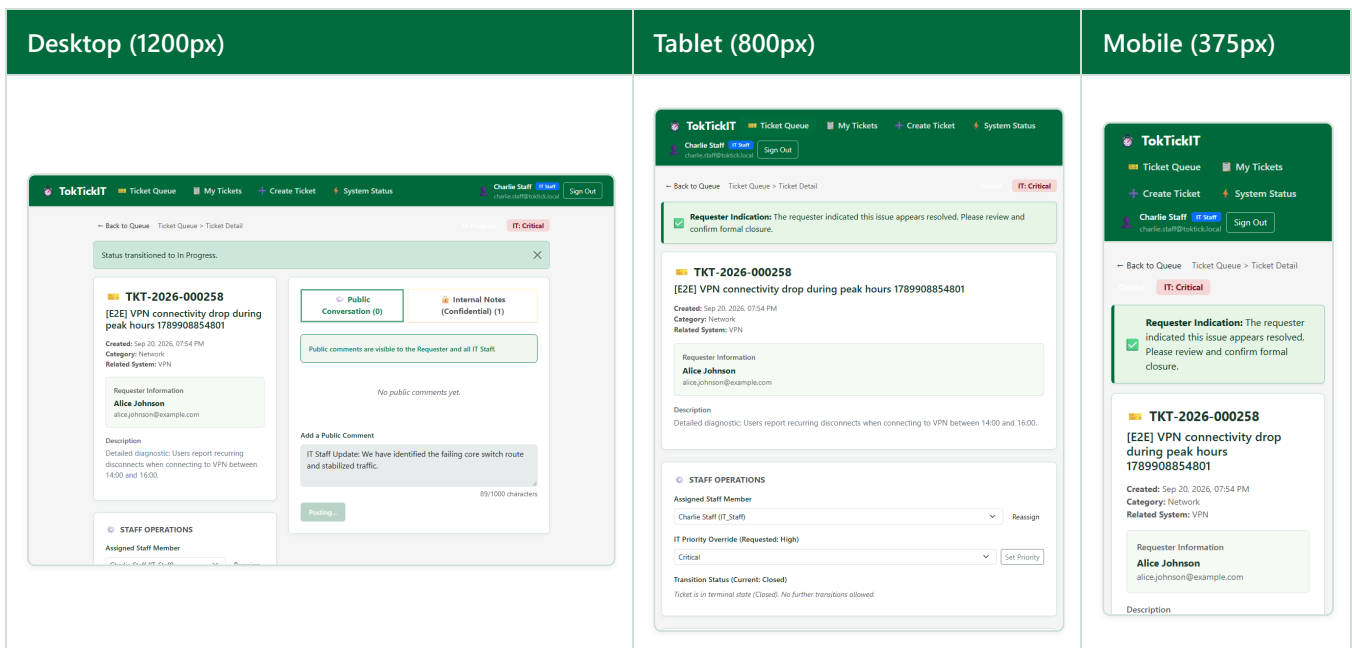
Login Screen Responsiveness:

Desktop (1200px)	Tablet (800px)	Mobile (375px)

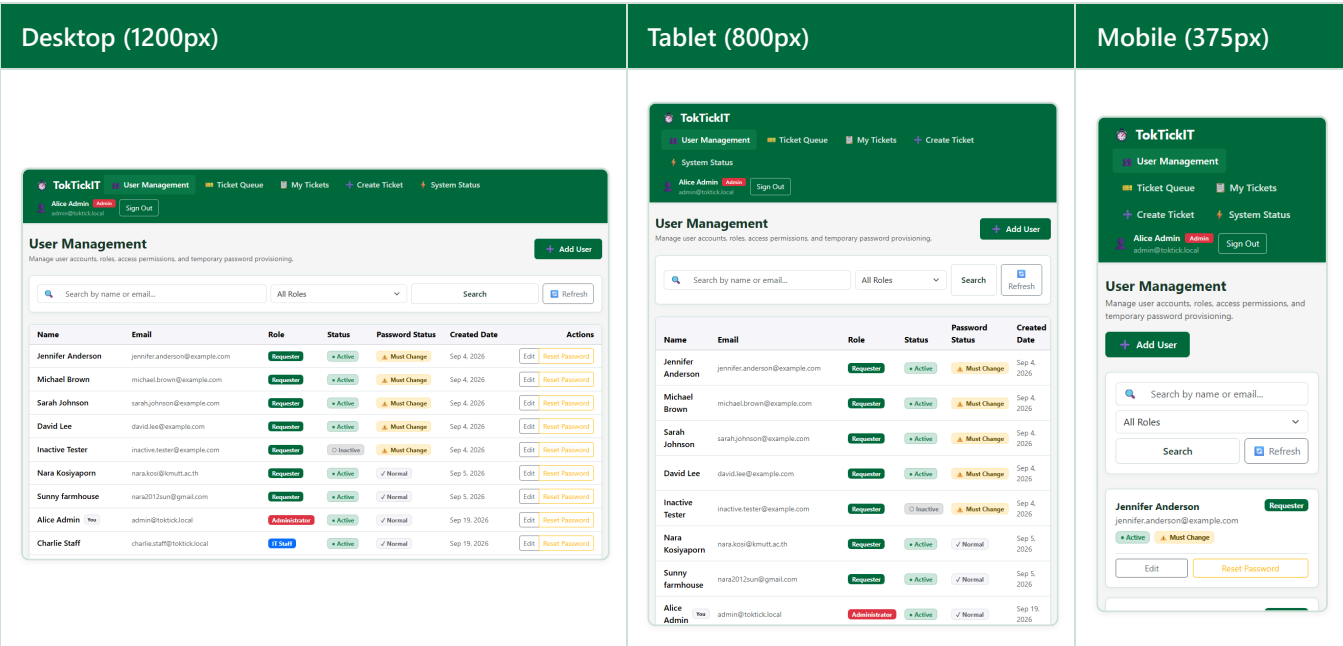
Staff Queue Responsiveness:



Staff Ticket Detail Responsiveness:



User Management Responsiveness:



9.2 Completed Visual Inspection Checklist

Checkpoint	Requirement	Verified Result	Status
Design Consistency	Reuse Zen Green color tokens, typography, and card elevation	Clean visual alignment across all screens with zero generic CSS	PASS
Role Navigation	Requesters see My Tickets & Create Ticket; Staff see Queue & Detail; Admin sees User Management	AppShell dynamically renders permitted tabs based on decoded JWT token role	PASS
Badges & Indicators	Distinct status pills, priority colors, and amber confidential banner	Status badges (New, In_Progress, Resolved, Closed) and priority tags formatted consistently	PASS
Editable vs. Read-Only	Editable inputs clearly bordered; read-only metadata styled with muted backgrounds	Uneditable fields (Ticket Number, Created Date, Requester Priority) styled with muted readonly styling	PASS
Validation Placement	Inline field errors and top alert banners	Complexity checklist on password change; modal error alerts on duplicate emails	PASS
Focus & Keyboard Navigation	Focus rings visible on buttons, inputs, and tab navigation	Standard :focus-visible styling applied across all interactive elements	PASS
Zero Horizontal Scroll	<code>scrollWidth <= clientWidth</code> on 375px mobile viewport	Verified by Playwright automated assertions (TC-E2E-AUTH-05, TC-E2E-STAFF-02, TC-E2E-ADMIN-05)	PASS
Clipping & Overlap	Tables transition to stacked cards on small screens; modal buttons wrap cleanly	No clipped text, overlapping elements, or broken layouts on any viewport	PASS